Probability distributions CHEAT SHEET

PROBABILITY

sample space S: all outcomes · 3 parts pass/fail: $2 \times 2 \times 2 = 8$

n choose x $= \frac{n!}{x!(n-x)!}$ 10 choose 3 = 120 · ordered: 720

assign: classical · relative frequency (38/200 = 0.19) · subjective

$0 \le P(A) \le 1$ · the outcomes' P sum to 1 · $P(\text{not A}) = 1 - P(A)$

$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$

app or newsletter: $0.6 + 0.4 - 0.25 = 0.75$

mutually exclusive: P(A and B) = 0, so just add

$P(A \mid B) = \frac{P(A \text{ and } B)}{P(B)}$ default | grade C: $\frac{0.024}{0.15} = 0.16$

$P(A \mid B) \neq P(B \mid A)$: $P(C \mid \text{default}) = 24/55 \approx 0.44$

independent: $P(A \mid B) = P(A)$, so $P(A \text{ and } B) = P(A)P(B)$

Bayes: $P(A_1 \mid B) = \frac{P(A_1)P(B \mid A_1)}{P(A_1)P(B \mid A_1) + P(A_2)P(B \mid A_2)}$

fraud: $\frac{0.005 \times 0.98}{0.005 \times 0.98 + 0.995 \times 0.02} = \frac{490}{2480} \approx 0.198$

RANDOM VARIABLES AND PORTFOLIOS

$f(x) \ge 0 \cdot \Sigma f(x) = 1$ · discrete uniform $f(x) = 1/n$

$E(X) = \mu = \Sigma x\, f(x)$ laptops: 2.15 a day

$\text{Var}(X) = \sigma^2 = \Sigma(x-\mu)^2 f(x) = E(X^2) - \mu^2$: 6.05 − 4.6225 = 1.4275

$E(aX + b) = aE(X) + b$ $\text{Var}(aX + b) = a^2\text{Var}(X)$

120X − 100: \$158, $\sigma = 120 \times 1.19 \approx$ \$143

$\sigma_{xy} = \Sigma\Sigma(x-\mu_x)(y-\mu_y) f(x,y)$ funds: −20.4

check: $\sigma_{xy} = [\text{Var}(X+Y) - \text{Var}\,X - \text{Var}\,Y]/2$

$\rho_{xy} = \frac{\sigma xy}{\sigma x\, \sigma y} = \frac{-20.4}{14 \times 2.96} \approx -0.49$

$R = wX + (1-w)Y$: $E(R) = wE(X) + (1-w)E(Y)$

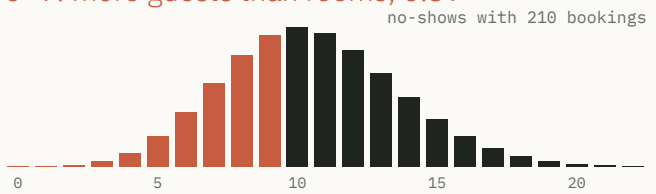
$\text{Var}(R) = w^2{\sigma_x}^2 + (1-w)^2{\sigma_y}^2 + 2w(1-w)\sigma_{xy}$

50/50: 9.1 %, $\sigma \approx 6.40$ % (the average σ: 8.48 %)

20/80: 7.36 %, $\sigma \approx 2.63$ %, below bonds alone (2.96 %)

BINOMIAL · POISSON · HYPERGEOMETRIC

0–9: more guests than rooms, 0.39



binomial: n independent trials, success p each time

$f(x) = (n \text{ choose } x)\, p^x (1-p)^{n-x}$ $\mu = np$, $\sigma^2 = np(1-p)$

10 calls, p 0.3: f(3) = 0.2668, $P(X \le 3) = 0.6496$

=BINOM.DIST(x, n, p, FALSE: exactly / TRUE: at most)

hotel, 210 bookings, p 0.05: $P(\text{no-shows} \le 9) = 0.3926$

Poisson: occurrences in an interval, $\mu = \sigma^2$ (rescale μ)

$f(x) = \mu^x e^{-\mu} / x!$ $\mu = 3$ an hour: f(5) = 0.1008

15 minutes: $\mu = 0.75$, $P(0) = e^{-0.75} \approx 0.47$

=POISSON.DIST(x, μ, FALSE / TRUE) · ≈ binomial, $\mu = np$

hypergeometric: n drawn without replacement from N, r successes

$f(x) = \frac{(r \text{ choose } x)(N-r \text{ choose } n-x)}{(N \text{ choose } n)}$ $\mu = n\frac{r}{N}$

$\sigma^2 = n\frac{r}{N}(1-\frac{r}{N})\frac{N-n}{N-1}$ 20 invoices, 5 wrong, 4 drawn:

f(1) = 2275/4845 ≈ 0.4696 (binomial: 0.4219) · $\sigma^2 \approx 0.63$

=HYPGEOM.DIST(x, n, r, N, FALSE) · sample < 5 % of N: binomial ok

Probability distributions CHEAT SHEET

UNIFORM AND NORMAL

shaded: P(X > $100) = 0.1056

continuous: $P(X = x) = 0$; probability = area under f(x)

uniform on [a, b]: $f(x) = \frac{1}{b-a}$ $\mu = \frac{a+b}{2}$ $\sigma^2 = \frac{(b-a)^2}{12}$

20–32 min: $P(X \le 25) = 5/12 \approx 0.4167$, $\mu = 26$, $\sigma \approx 3.46$

normal: $f(x) = \frac{1}{\sigma\sqrt{(2\pi)}} e^{-(x-\mu)^2/(2\sigma^2)}$ symmetric about μ

within 1, 2, 3 σ: 68.3 %, 95.4 %, 99.7 %

$z = \frac{x-\mu}{\sigma}$ table: area left of z; $\Phi(1.25) = 0.8944$, $\Phi(1) = 0.8413$

above: $1 - \Phi(z)$ · between: $\Phi(b) - \Phi(a)$

$\Phi(1.25) - \Phi(-0.5) = 0.8944 - 0.3085 = 0.5859$

orders μ \$80, σ \$16: $P(X > 100) = 1 - 0.8944 = 0.1056$

top 5 %: $x = \mu + z\sigma = 80 + 1.645 \times 16 \approx \$106.32$

=NORM.DIST(x, μ, σ, TRUE) · =NORM.INV(area, μ, σ)

=NORM.S.DIST(z, TRUE) · =NORM.S.INV(area): 0.95 → 1.645

NORMAL APPROXIMATION OF THE BINOMIAL

binomial ≈ normal if $np \ge 5$ and $n(1-p) \ge 5$

$\mu = np$, $\sigma = \sqrt{(np(1-p))}$

continuity correction: $P(X \le 25) \approx P(Y \le 25.5)$

n 100, p 0.3: $z = (25.5 - 30)/4.58 \approx -0.98 \rightarrow 0.16$

exact binomial: 0.1631

without it: $z \approx -1.09 \rightarrow 0.14$ (too low)

EXPONENTIAL

exponential (time between events): $f(x) = \frac{1}{\mu} e^{-x/\mu}$

$P(X \le x) = 1 - e^{-x/\mu}$ mean = $\sigma = \mu$ · no memory

tickets every 20 min on average: $P(\le 10) = 1 - e^{-0.5} \approx 0.39$

=EXPON.DIST(x, 1/μ, TRUE): the rate, not the mean

Poisson counts ↔ exponential gaps: 3 an hour ↔ 20 min;

$P(0 \text{ in } 30 \text{ min}) = e^{-1.5} = P(\text{gap} > 30) \approx 0.2231$

WHICH DISTRIBUTION?

successes in n independent trials → binomial

occurrences in an interval of time or space → Poisson

without replacement from a small group → hypergeometric

equally likely over a range → uniform

a bell-shaped measurement → normal

time between random events → exponential

THE CASE, IN SIX LINES

Halverton Hotels (fictional · our own data): 200 rooms

250 sold-out nights: no-shows mean 10, variance 9.502

binomial(200, 0.05): $np(1-p) = 9.5$ · it fits

overbook by the mean (210): guests > rooms on 39 % of nights

$\text{gain}(k) = 160 \times E[\text{filled} - 190] - 450 \times E[\text{walked}]$

best: 208 bookings, +\$997 a night; walk on ≈ 18 % of nights

SPREADSHEET FUNCTIONS

COMBIN · PERMUT · SUMPRODUCT · SQRT · EXP · LN

BINOM.DIST · POISSON.DIST · HYPGEOM.DIST

NORM.DIST · NORM.INV · NORM.S.DIST · NORM.S.INV

STANDARDIZE · EXPON.DIST · AVERAGE · VAR.S